THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NJOMBE CITY COUNCIL

FORM FOUR PRE -MOCK EXAMINATION APRIL -2024

032/1 **CHEMISTRY 1**

TIME: 3:00 Hours **18TH April 2024**

INSTRUCTIONS

1. This paper consists of sections A, B and C with a total of eleven (11) questions.
2. Answer all questions in section A and B and only two questions in section C
3. Any unauthorized materials are not allowed in the examination room
4. Write your examination number on every page of your answer book sheets
5. The following constant may be used:

Atomic masses: H=1, Cl=35.5, K=39, O=16, Na=23, Ba=137, N=14 Al=27, S=32, Pb=207, Na= 23, C=12, H=1, K=39, Mn= 55, Mg=24, Zn=65

- Avogadro's number = 6.02 × 1023Particles
- 1 Faraday = 96500C
- Standard Pressure = 760mmHg
- Standard Temperature = 273K
- 1litre= 1dm3=1000cm3=1000ml
- GMV at s.t.p =22.4dm

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SECTION A (16 Marks)
Answer all questions from this section

1. For given alternatives below choose the most correct answer from (i)-(x) and write its answer in the booklet or answer sheet provided;

   i. In an experiment a form three student was in need of measuring 200cm^3 of water quickly he would use what type of apparatus below to obtain exact amount needed for the experiment to continue;

   A. A conical flask
   B. A round-bottomed flask
   C. A measuring cylinders
   D. A test-tube
   E. A trough

   ii. People on the universe tend to destroy the environment due to man-made activities including burning elements in oxygen, which group from the alternatives will result to the formation of acidic oxide?

   A. Sodium, magnesium and carbon
   B. Sulphur, carbon and phosphorous
   C. Phosphorous, carbon and calcium
   D. Sulphur, carbon and potassium
   E. Oxygen, nitrogen and boron

   iii. When a teacher was teaching, he explained that copper oxide always changes to copper metal using hydrogen and the asked them, what does that process mean? They failed assist them from the following to obtain the correct answer

   A. Copper oxide is oxidized to copper metal
   B. Hydrogen is reduced to water
   C. Hydrogen is an oxidizing agent
   D. Hydrogen reduced copper oxide
   E. Hydrogen is a catalyst

   iv. When dil nitric acid and silver nitrate were added to an original soluble salt, a white precipitate was formed. This experiment simply say that the original salt must have

   A. Nitrate
   B. Carbonate
   C. Sulphate
   D. Chloride
   E. Hydroxide

   v. Juma was around the special building where he found a certain chemical compound with the chemical formula of $ZnSO_4$. $7H_2O$.From your knowledge what does this signifies in terms of a molar mass of it?

   A. 289.
   B. 287.
   C. 342
   D. 453
   E. 286

   vi. Form three student were conducting the experiment about how magnesium can produce the so-called Faraday what do you think will be number of Faradays that are formed from Magnesium ion during their experiment from the following

   A. 4.
   B. 6
   C. 2.
   D. 7.
   E. 12

   vii. Tanzania with the collaboration of China started the negation on how they could obtain iron from its ore using a blast furnance. As an expert on extraction of that metal. What type of materials that will enter into the furnance from the following in order to obtain a pure metal

   A. Iron, Limestone and Coke.
   B. Iron ore, coke and slag
   C. Coke, Limestone and iron ore
   D. Water, Carbon dioxide and limestone.
   E. Iron ore, Coke and quicklime.

viii. One form one student entered in laboratory and accidentally dropped a yellow-coloured flowers in a gas jar containing Chlorine gas. What will be observed from the following

A. Flowers will dry
B. Flowers will change colour to pink
C. There will be decolorization of flowers
D. Flowers will shrink
E. The flower will turn blue

ix. A and B are elements in the same period of the periodic Table. A is in group IIand B is in group III. Which of the following statements is not true about thetwo elements?

A. B has one proton more than A in the nucleus
B. The atomic number of B is one unit greater than that of A
C. A has one electron less than B in its outermost shell
D. A has one electron more than B in its outermost shell
E. A and B have the same number of shells

x. If two gas jars labeled A and B contains 22.4dm3 of oxygen and 22.4dm3 of nitrogen respectively at STP, then it is true that

A. There were 6.02×10^{23} molecules of oxygen in A and 6.02×10^{23} molecules of nitrogen in B
B. There were 6.02×10^{23} atoms of oxygen in A and 6.02×10^{23}atoms of nitrogen in B
C. There were 12.04×10^{23} molecules of oxygen and nitrogen in their respective jars
D. 6.02×10^{23} molecules of oxygen in A and 6.02×10^{23} molecules of nitrogen in jar A and B respectively
E. Jars A and B contains 96,500 molecules of oxygen and nitrogen respectively

2. Match the items in list A with response in list B and write its corresponding correct letter to the answer book provided

List A		List B
i.	Oxides of lead, zinc and aluminum	A. Carbon
ii.	Its oxide is black when hot and when cool	B. Acidic Oxides
iii.	Its oxide is a drying agent in Haber process	C. Amphoteric oxides
iv.	Strong oxidizing agent	D. Sodium
v.	Weak reducing agent	E. Zinc Oxide
vi.	Its oxide dissolves in water	F. Calcium
		G. Copper
		H. Potassium permanganate
		I. Boron

SECTION B (54 marks)

Answer all questions from this section

3.

a) Simon put some iron wool into two bottle tops, she heated one of the samples of iron wool from above using a blow pipe. The iron wool glown and the heated and wool looked darker in color and stiffer than the unheated iron wool. Explain why so happened.

b) A form three needed to know the reasons for what happened to the questions below assist them to explain based on the following observations,
   i. A fire associated with electrical equipment's is not given a full class.
   ii. Iron (ii) Chloride is rarely formed.
   iii. Concentrated sulphuric acid do not conduct electricity but dilute sulphuric acid does.

4.

a) There was the debate between you and form two about the structure of an atom. To end the debate and to be a winner of it, draw a well labeled diagram which show the structure of an atom.
b) Proton and Neutron are both nucleon but in fact are separable in their properties. Write three points showing their dissimilarities.

5.

a) The reddish-brown oxide of metal M is formed by the action of moisture and Oxygen on the metal surface.
   i. Identify metal M
   ii. What are the three conditions necessary for the formation of reddish brown on metal surface?
   iii. One Student from Luponde Secondary School failed to suggest the methods for preventing what occurred on the surface of metal M. Please help that student

b)
   i. B and D has contradictions on meaning of ISOTOPY and ISOTOPES. Briefly explain the meaning of each term as you help B and D to solve the contradictory effect they have.
   ii. The isotopes of a certain element are in the ratio of 3:2:1 and their Molar Masses are 16, 17 and 18 respectively. Calculate the relative atomic mass of an element

6.

a) An employer wanted to produce more copper because it's required in the market in large quantities, he decided to use the electrolysis method for exploitation, but he didn't know the factors to consider so that he can produce more copper. As a young scientist of this century with enough knowledge of electrolysis. Explain three factors on which the amount of copper deposited will depend on

b) If the amount of copper that was required to be exported by the employer was 12kg.How will you calculate the number of Amperes that would be used for 2hours

7.

a) During holidays your two young brother was given a gift of 5g and 8g of calcium carbonate. Then when they a lived at home they stored in the same container, next day in the morning your young sister accidentally poured dilute hydrochloric acid in that container and a colourless gas **X** was produced , what is the name of that gas?
b) What volume of the gas **X** was produced
   i. What chemical test will you conduct to identify gas **X**
c) If the gas was collected by your young siter, how would she use the gas

8.

a) Suppose you have been instructed by your chemistry teacher to mention the general components of soil at your school. What five components of the soil will you consider?
b) What is the importance of soil texture to the growth of plant?

SECTION C (30 marks)
Answer only two questions

9. Suppose you were in a study tour at Roman boys' secondary school and eventually the form five invited you to explain on how sulphur can be extracted from the ground. With the aid of diagram, explain how you can apply the knowledge you have to obtain sulphur from the ground.

10. Many countries this year have experienced shortage of rainfall that has resulted to poor yield of crops and death of animal due to lack of water and pastures. Scientists have advised that the problem is due to air pollution in many countries in the world. Explain what might happen in the future to the world if the present trend of polluting the air is not controlled and checked. (Six points)

11. Contact process deals with manufacture of sulphuric acid where a number of stages are involved with different equations, but one chemical equation looks very important than all others.
    a) Write chemical equation for this industrial important reaction.
    b) By using Le-chatellier's principle, explain what should be done to obtain high yield of the product.
    c) Explain what will happen if the product is removed from a system.
    d) Name a catalyst used in contact process using its IUPAC name and write its chemical formula.
    e)
        i. Name and write an equation for another industrial important process you studied in Energetics rather than Contact process.
        ii. Draw the energy profiles diagrams for exothermic reaction and endothermic reaction respectively.